

D.2 Worker Adoption and Reported Effects

D.2.1 Regression Specification

Next, we examine how workers’ adoption of and benefits from AI chatbots vary with the employer-led chatbot initiatives. Specifically, we compare workers in the same occupations with similar characteristics and estimate how their adoption behaviors and reported benefits vary with employer initiatives for AI chatbots:

$$Y_i = \gamma' X_i + \beta' \text{EmployerInitiatives}_i + \varepsilon_i, \quad (69)$$

where Y_i is a chatbot-related outcome for worker i , such as use or reported benefits of AI chatbots for work; X_i is a vector of worker pre-determined characteristics, including age, gender, experience, and occupation fixed effects;²⁹ and $\text{EmployerInitiatives}_i$ is a vector of employer-driven initiatives, including encouraging employees to use AI chatbots, investing in enterprise chatbot solutions, providing employee training— and combinations thereof. We estimate Equation (69) by OLS and cluster the standard errors by workplace-occupation level.

The specification in Equation (69) is motivated by the literature on “Productivity J-Curves,” which emphasizes that complementary organizational investments may be critical for realizing the gains from new technologies (Brynjolfsson and Hitt, 2000; Brynjolfsson, Rock and Syverson, 2021). Moreover, because early adoption of AI chatbots was largely worker-driven, this analysis sheds light on the extent to which employers can move the needle and influence outcomes by also investing in the technology. Appendix A.1 presents a Roy-style model of chatbot adoption to help interpret the estimates in Equation (69). We draw on this framework in the discussions that follow.

Although employer initiatives are not assigned through a randomized design, our rich data on firms and workers allow us to examine the robustness of our findings to several potential confounders. Appendix E.4 presents these robustness checks. First, we show that all results hold when controlling for firm characteristics (from Table D.1),

²⁹All worker characteristics are measured in 2022, before AI chatbots became available.

ensuring that differences across employer initiatives are not driven by variation in firm age, size, or productivity. Second, we show that the results are robust to controlling for workers’ detailed task mixes within occupations, demonstrating that the effects of employer initiatives are not driven by differences in the types of tasks more amenable to AI chatbot use.³⁰ Finally, in our panel analyses in Section 3, we verify that adopters of different chatbot initiatives followed similar labor market trajectories prior to the arrival of AI chatbots. Together, these robustness checks help rule out confounding factors and support the validity of our estimated relationships.

Finally, since our data are collected from workers rather than senior leadership, our estimates reflect employer initiatives *as perceived by workers*. Since employer policies may not be perfectly communicated, worker self-reports may not fully align with employer intentions. Nonetheless, Appendix E.5 shows that our findings are largely robust to using coworker reports to measure employer initiatives, demonstrating these may be interpreted as workplace-wide treatments.

D.2.2 Adoption

Figure D.3 shows how AI chatbot adoption varies with employer initiatives for a typical worker. The top estimate (*No Initiative*) highlights the bottom-up nature of adoption: even among workers whose employers neither encourage chatbot use, provide an enterprise chatbot, nor offer training, about 41% have used AI chatbots at work at some point, and 7% use them daily.

Despite this high baseline, employer initiatives are associated with substantially higher adoption. Implemented individually, *Encouraged use*, *Own chatbot*, and *Training* each raise extensive-margin take-up rates to 76%, 57%, and 69%, respectively. Encouraged-use policies are especially effective at increasing intensive usage, raising daily use from 7% to 18%. In contrast, providing training or an enterprise chatbot without encouragement

³⁰We do not include task controls in our main specification, as task mix may itself be endogenous to chatbot adoption. Indeed, Section E.1.6 shows that AI chatbots have created new job tasks for many workers.

leads to only modest increases in daily use, to 9% and 8%, respectively.

The bottom rows show adoption rates when employer initiatives are combined. Adoption peaks when all three—encouragement, enterprise chatbots, and training—are in place: 93% of workers in such settings report having used AI chatbots at work, and 28% use them daily.

Employer initiatives not only boost adoption—they also influence *who* adopts the tools. Figure D.4 examines how the gender gap in AI chatbot adoption varies with employer initiatives, controlling for workers’ occupation, age, experience, earnings, and wealth.

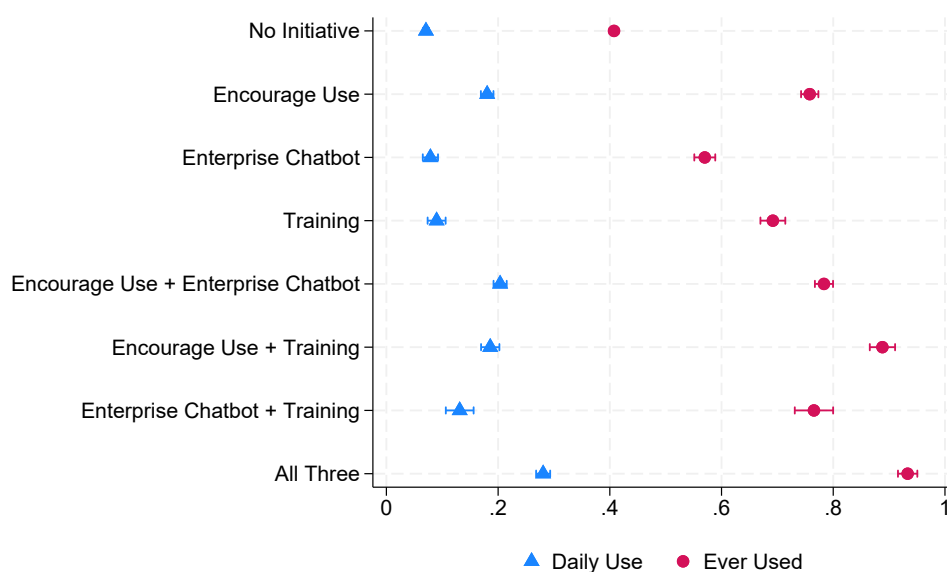
The *No Initiative* estimate shows that a substantial gender gap has emerged in chatbot adoption: without encouraged use, enterprise chatbots, or training, women are about 13 percentage points less likely than comparable men in the same occupation to have used AI chatbots.³¹

Notably, employer initiatives can significantly reduce this gap, with training proving especially effective. For instance, providing training alone narrows the gap to 7 percentage points, and combining training with other initiatives reduces it further, to just 2–3 percentage points.

Table E.3 analyzes how employer initiatives affect other demographic gaps in adoption, and Appendix Figure E.5 shows gender gaps in daily usage. Employer initiatives also help close gaps by age, experience, and earnings. Training, in particular, is consistently effective in narrowing gaps in daily use as well. By contrast, encourage-use policies alone *widen* the gender gap in daily usage—from 3 to 10 percentage points. However, when encouragement is paired with training, the gap reverses: women become *more* likely than men to use AI chatbots daily, with a gender gap of –2 percentage points.

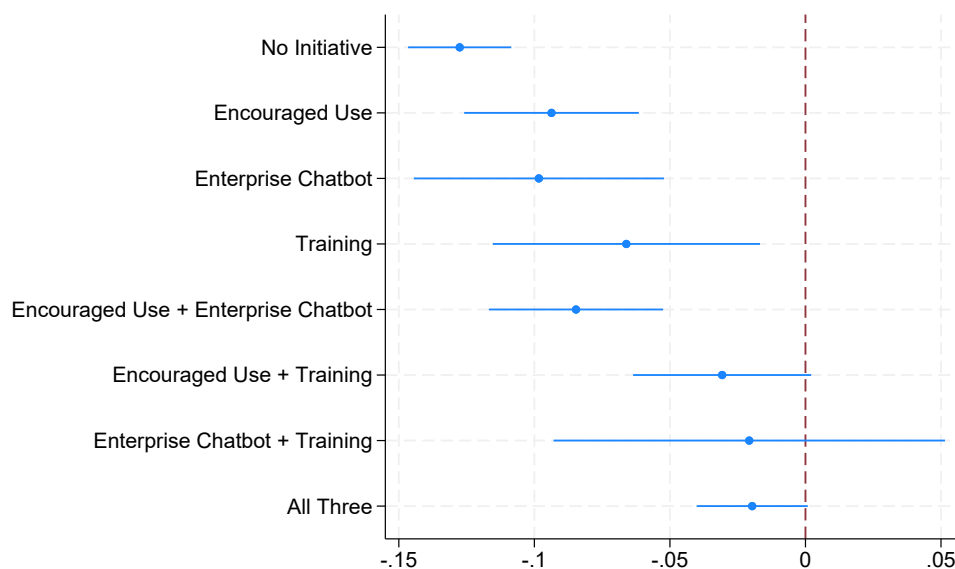
³¹Figure E.4 shows that the gender gap in chatbot adoption persists even in workplaces where the tools are prohibited. Although employer bans are associated with a reduction in overall adoption to roughly one-third of the baseline rate, about 80% of the baseline gender gap remains. This pattern mirrors findings by Carvajal, Franco and Isaksson (2024), who document a similar gender gap in ChatGPT use among Norwegian bachelor’s students and show, using a vignette experiment, that the gap largely reflects male students continuing to use the tools even when explicitly banned.

Figure D.3: Employer Initiatives and Chatbot Adoption



Notes: This figure illustrates how workers' use of AI chatbots vary with their employer initiatives. The estimates are based on predicted values from Equation (69), varying EmployerInitiatives while holding workers' characteristics X at their mean values. Whiskers represent 95% confidence bands of the predicted values. Some confidence intervals are too narrow to be distinguished from the point estimates at this scale. Table E.2 reports the full set of regression estimates, including specifications that use the share of workers with *Monthly or More* and *Weekly or More* usage as dependent variables. *Sample:* All completed responses from the 2024 survey linked to registry data.

Figure D.4: Employer Initiatives and Gender Gaps in Adoption



Notes: This figure illustrates how gender gaps in AI chatbot adoption varies with employer initiatives. The estimates are obtained from a multivariate regression of AI chatbot adoption on worker characteristics X , controlling for occupation fixed effects, and are estimated separately based on employers' AI chatbot initiatives. Worker characteristics include gender, age, experience, earnings, and net wealth; see Table E.3 for the full set of regression estimates. Whiskers represent 95% confidence intervals. *Sample:* All completed responses from the 2024 survey linked to registry data.